Summary in Graph

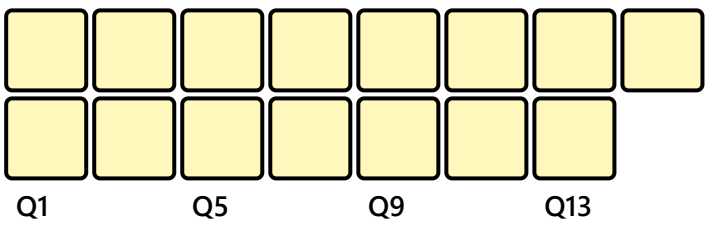
Exam Summary (GO Classes GATE CS | Propositional Logic | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (7 + 8)
Total Marks:	23 (7 + 16)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (147) out of 1465 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Mathematical Logic

If $\mathbf{p}$ is true, $\mathbf{q}$ is true, and $\mathbf{r}$ is true, find the truth value of the statement.

$$(p \wedge q) \leftrightarrow (q \vee \sim r)$$

Choose the correct answer below.

A. True because $(p \wedge q)$ is true and $(q \vee \sim r)$ is true.
B. False because $(p \wedge q)$ is false and $(q \vee \sim r)$ is false.
C. True because $(p \wedge q)$ is false and $(q \vee \sim r)$ is false.
D. False because $(p \wedge q)$ is true and $(q \vee \sim r)$ is true.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

If p is true and q is false then the truth values of $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ and $(\sim p \vee \sim q) \wedge (\sim q \vee p)$ are respectively

A. True, True
B. True, False
C. False, False
D. False, True

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

If $(p \wedge \sim q) \wedge (p \wedge r) \rightarrow \sim p \vee q$ is false, then the truth values of p, q and r are, respectively :

A. F, T, F
B. T, F, T
C. T, T, T
D. F, F, F

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

If p, q, r are simple statement with truth values T, F, T respectively then the truth value of $((\sim p \vee q) \wedge r) \rightarrow p$ is :

A. True
B. False
C. True if r is false
D. True if q is true

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job." Which of the following English Statement expresses the statement $p \rightarrow q$?

A. "If Maria learns discrete mathematics, then she will find a good job."
B. "Maria will find a good job provided that she learns discrete mathematics."
C. "Maria will learn discrete mathematics provided that she finds a good job."
D. "For Maria to get a good job, it is sufficient for her to learn discrete mathematics."

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 03sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

A sufficient condition for a triangle T be a right triangle is that $a^2 + b^2 = c^2$. An equivalent statement is

A. If T is a right triangle then $a^2 + b^2 = c^2$.
B. If $a^2 + b^2 = c^2$ then T is a right triangle.
C. If $a^2 + b^2 \neq c^2$ then T is not a right triangle.
D. T is a right triangle only if $a^2 + b^2 = c^2$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Mathematical Logic

Suppose that the statement $p \rightarrow \neg q$ is false. What is the number of all possible combinations of truth values of r and s for which $(\neg q \rightarrow r) \wedge (\neg p \vee s)$ is true?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

A very special island, "Smullyan's island", is inhabited only by knights and knaves. Knights always tell the truth, and knaves always lie. You encounter two people, A and B. A says "The two of us are both knights" and B says "A is a knave".
Determine what A and B are, respectively, if they address you in the above way described:

A. Knight, Knight
B. Knight, Knave
C. Knave, Knight
D. Knave, Knave

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Given the truth table of a Binary Operation $ as follows:

X	Y	X$Y
1	0	1
1	1	1
0	1	0
0	0	1

Identify the matching Boolean Expression.

A. $X\$\neg Y$
B. $\neg X\$Y$
C. $\neg X\$\neg Y$
D. none of the options

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars.
Determine the truth value of each of these propositions for the most recent fiscal year. Which is/are True?

A. Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.
B. Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.
C. Acme Computer had the largest net profit or Quixote Media had the largest net profit.
D. If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let P, S, R be three statements(propositions). Let S be a sufficient condition for P, Let R is a necessary condition for P then which of the following is/are true?

A. S is a sufficient condition for R.
B. S is a necessary condition for R.
C. S is neither sufficient, nor a necessary condition for R.
D. S is a sufficient and necessary condition for R.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Consider the following popular puzzle.

When asked for the ages of her three children, Mrs. Baker says that “Alice is her youngest child if Bill is not her youngest child, and that Alice is not her youngest child if Carl is not her youngest child.” Note that only one of the three children can be her youngest child.

Which is correct?

A. Alice is her youngest child.
B. Bill is her youngest child.
C. Carl is her youngest child.
D. Information is not sufficient to find out the youngest child.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Consider the following popular puzzle.

A boy and a girl are talking. One of them has black hair, another has white hair.
"I am a boy" said the child with black hair.
"I am a girl" said the child with white hair.
At least one of them is lying.

Which of them is lying?

A. The boy only
B. The girl only
C. Both of them
D. Information is not sufficient to find out the liar

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let P be a compound proposition over 4 propositional variables: a, b, c, d. We know that for a compound proposition over n propositional variables, we have 2^n rows in the truth table. Every row of truth table of P is called an Interpretation of P. A row in truth table of P is called "model" iff P is true for that row. Let P be $'a \rightarrow b'$. How many models are there for P?

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

The number of combinations of truth values for p, q and r for which the statement $\neg p \leftrightarrow (q \wedge \neg(p \rightarrow r))$ is true _____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

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